

RE-RUN: THE CAMPUS RUNNER

Comprehensive Game Release & Technical Build Specification Document

1. Build Identification & System Release Metadata

Build Identifier:	RE-RUN-v1.0.0-STABLE-GODOT2D	Release Target:	Windows x64 / Godot 4.3
GitHub Repo:	https://github.com/stewin16/RE-RUN	Repository Branch:	master (Clean HEAD)
Launcher Script:	PLAY_GAME.bat (One-Click)	Executable:	Godot_v4.3-stable_win64_console.exe
Asset Packages:	KW_School_Character_Set_Free.zip	Sky Package:	pixel_art_clouds.zip

2. System Requirements & Hardware Benchmarks

Specification	Minimum Requirement	Recommended Specification
Operating System	Windows 10 / 11 (64-bit)	Windows 11 (64-bit)
Processor (CPU)	Dual-Core 2.0 GHz x86-64	Quad-Core 3.0 GHz x86-64 or better
Memory (RAM)	2 GB RAM	4 GB RAM or higher
Graphics (GPU)	OpenGL 3.3 / Direct3D 11 compatible	Dedicated GPU with Direct3D 12 support
Display Resolution	960 x 540 (Windowed / Scale 1x)	1920 x 1080 (Fullscreen / 60 FPS Target)

3. Visual Asset Showcase (Latest Build Captures)

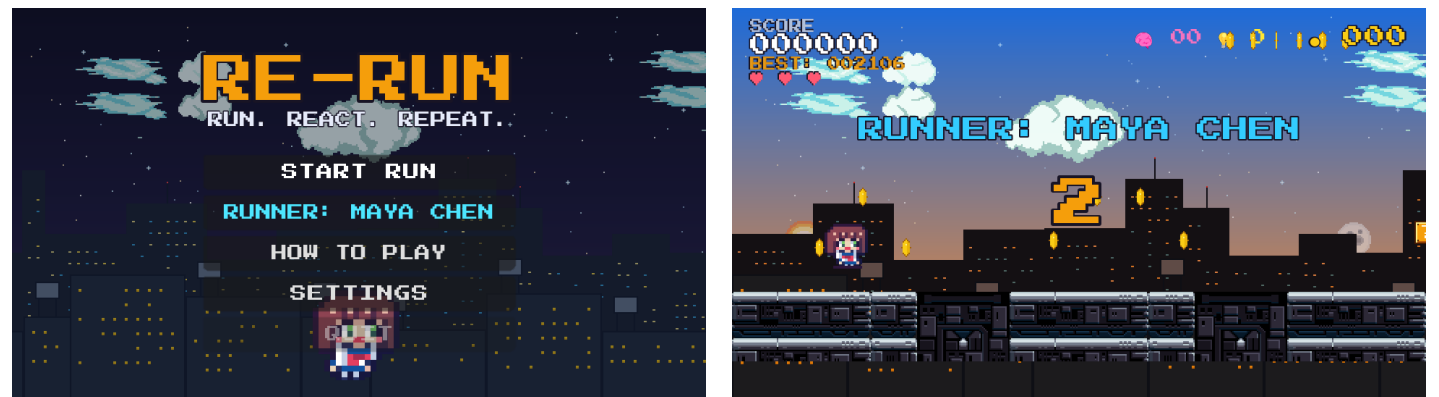


Figure 1: Main Menu & Student Roster Select (Left) | Parallax Sky & Pixel Art Cloud Drift (Right)

4. Controls, Inputs & Keybindings Specification

Action / Trigger	Primary Keybindings	State Machine Execution
Horizontal Run	A / D or Left / Right Arrows	Accelerates player velocity.x up to SPEED (280-420 px/s)
Ground Jump	Space / W / Up Arrow	Executes vertical launch (JUMP_FORCE = -390.0 px/s)
Mid-Air Double Jump	Space / W / Up Arrow (In Air)	Triggers secondary mid-air boost (DOUBLE_JUMP = -350.0 px/s)
Slide Under Pipes	S / Down Arrow	Shrinks hit box to 14x12px for 0.42s under clearance pipes
Slash Action Attack	J / F / Z / Left-Click	Activates 28x24px slash Area2D for 0.18s to break hazards
Pause / Quick Restart	P (Pause) R (Restart)	Toggles game pause modal or quick level restart

5. Physics & Boss Battle Demonstrations



Figure 2: Student Mid-Air Jump Execution (Left) | Department Professor Boss Battle (Right)

6. Academic Quiz Modal & Question Block System

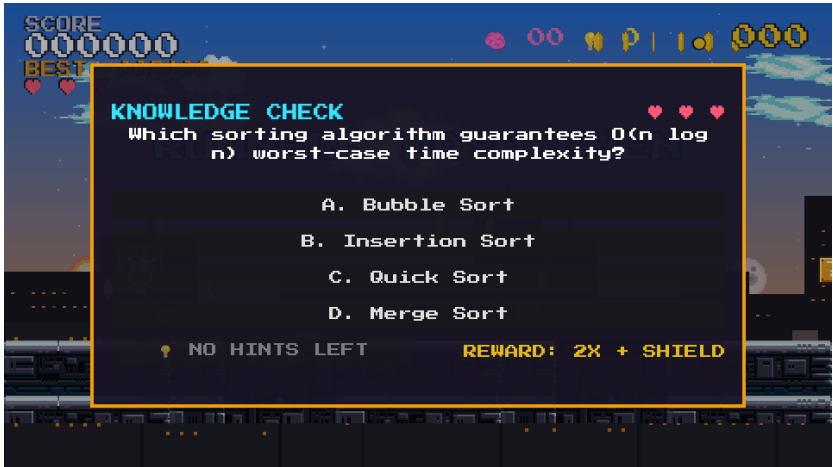


Figure 3: Mario-Style Question Block Quiz Interface (Interactive Multiple Choice Modal)

7. Complete Asset Manifest & Character Roster

All characters are sourced exclusively from KW_School_Character_Set_Free.zip. The project provides player selection liberty across **22 Student Characters**:

Category	Count	Character Identifiers & Major Descriptions
Male Students	11 Characters	Leo Tanaka (CS), Kai Sterling (Cybersec), Ren Takahashi (Game Dev), Jin Mori (DevOps), Arata Sato (Robotics), Daiki Kudo (Graphics), Haruto Kuroda (Compilers), Kaito Minami (Distributed), Sora Tachibana (Bioinformatics), Riku Nakajima (AI), Yuto Hayashi (Networking)
Female Students	11 Characters	Maya Lin (AI & Data Science), Chloe Vance (Software Eng), Aoi Sato (UI/UX Design), Hana Mori (Data Science), Yui Tanaka (Embedded Systems), Emi Kuroda (Cyber-Defense), Sakura Minami (Robotics), Rin Kudo (Game Physics), Mei Takahashi (Cloud Native), Nina Sterling (Compilers), Sora Lin (Quantum Comp)
Faculty Bosses	5 Teachers	Prof. Sterling (Level 1: Algorithms), Prof. Vance (Level 2: Cybersecurity), Prof. Kuroda (Level 3: Systems), Prof. Sato (Level 4: AI), Prof. Lin (Level 5: Quantum)

8. Project Directory & Scene Architecture Map

Directory Path	File / Subfolder	Architectural Responsibility
RE-RUN/RE-RUN/	project.godot	Main Godot project configuration & viewport settings
Scenes/Levels/	level_1.tscn	Main game level with procedural chunk spawner & sky parallax
Scenes/Player/	player.tscn	Character Body 2D, AnimatedSprite2D, slash collision, audio
Scenes/Objects/	teacher_boss.tscn	Boss node with levitation, attack states, flying exam paper
Scenes/UI/	main_menu.tscn, hud.tscn	Character select carousel, lifelines, score, quiz modals
Scripts/UI/	game_settings.gd	Global settings singleton storing character roster & state

9. Build Verification & One-Click Launch Instructions

- Clone repository: git clone <https://github.com/stewin16/RE-RUN.git>
- Run launcher: Double click PLAY_GAME.bat in the repository root directory.
- Manual CLI Launch: godot_bin\Godot_v4.3-stable_win64_console.exe --path RE-RUN\RE-RUN